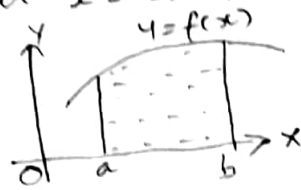


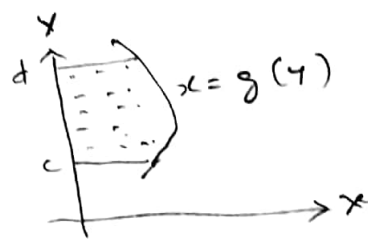
MODULE IIIAreas of Plane regions

Area bounded by the curve $y = f(x)$, the x -axis and the ordinates at $x = a$ and $x = b$ is given by

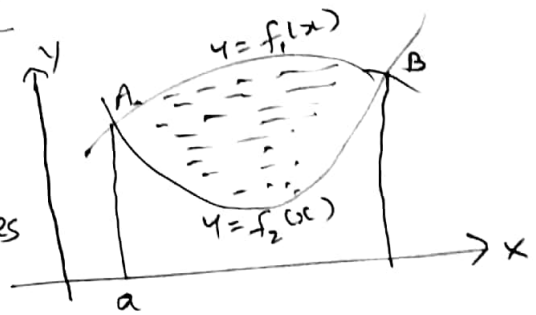
$$A = \int_a^b f(x) dx.$$



Note The area bounded by the curve $x = g(y)$, the y -axis and the abscissa at $y = c$ and $y = d$ is given by $A = \int_c^d g(y) dy$

Area between two curves

Let $y = f_1(x)$ and $y = f_2(x)$ intersect at A and B and a and b be the x -coordinates of the points of intersection.

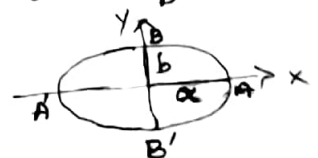


Area between these two curves is given by

$$\begin{aligned} A &= \int_a^b f_1(x) dx - \int_a^b f_2(x) dx \\ &= \int_a^b [f_1(x) - f_2(x)] dx. \end{aligned}$$

Examples

- (1) Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- Let $AA' = 2a$ be the major axis
 $BB' = 2b$ be the minor axis



since its eq contains only even degree terms in x and y , ellipse is symmetrical about both co-ordinate axes. Therefore these axes divide it into four parts (quadrants) whose areas are equal

$\therefore$ Area of the whole ellipse

= $4 \times$ its area in first quadrant

= $4 \times$ area of elliptic sector OAB

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = \frac{a^2 - x^2}{a^2}$$

$$y = \frac{b}{a} \sqrt{a^2 - x^2}. \text{ The points A and B}$$

are $(a, 0)$ and $(0, b)$ respectively

$$\begin{aligned} \text{area of OAB} &= \int_a^b y \, dx = \frac{b}{a} \int_a^b \sqrt{a^2 - x^2} \, dx \\ &= \frac{b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a \\ &= \frac{b}{a} \left[0 + \frac{a^2}{2} \sin^{-1}(1) - 0 \right] \\ &= \frac{b}{a} \frac{a^2}{2} \times \frac{\pi}{2} = \frac{\pi ab}{4} \end{aligned}$$

$$\therefore \text{Area of the whole ellipse} = 4 \times \frac{\pi ab}{4} = \pi ab.$$

(2) Find the area bounded by the curve $\sqrt{x} + \sqrt{y} = 1$ and the x -axis and y -axis

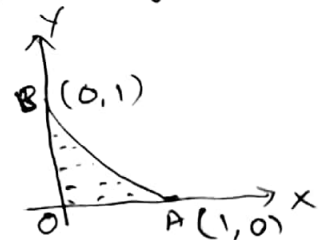
$$\sqrt{x} + \sqrt{y} = 1 \longrightarrow (1)$$

Intersection of the curve with x -axis is got by putting

$$y = 0 \quad \therefore \text{we get } \sqrt{x} = 1 \text{ i.e. } x = 1$$

Intersection of the curve with y -axis is got by putting $x = 0 \therefore$ we get $\sqrt{y} = 1$ i.e. $y = 1$

$\therefore$ A is $(1, 0)$ and B $(0, 1)$



$$\begin{aligned}\sqrt{x} + \sqrt{y} &= 1 \Rightarrow \sqrt{y} = 1 - \sqrt{x} \\ \Rightarrow y &= (1 - \sqrt{x})^2 \\ \Rightarrow y &= 1 - 2\sqrt{x} + x^2\end{aligned}$$

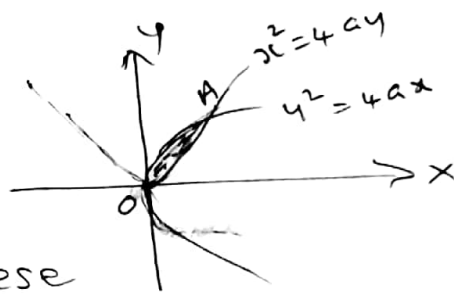
Required area is given by

$$\begin{aligned}A &= \int_0^1 y \, dx = \int_0^1 (1 - 2\sqrt{x} + x^2) \, dx \\ &= \int_0^1 (1 - 2x^{1/2} + x^2) \, dx \\ &= \left(x - 2 \frac{x^{3/2}}{3/2} + \frac{x^3}{3} \right)_0^1 \\ &= 1 - \frac{4}{3} + \frac{1}{3} \\ &= \frac{6 - 4 + 1}{6} = \frac{1}{6} \text{ sq. units.}\end{aligned}$$

(3) Find the area common to the parabolas

$$y^2 = 4ax \text{ and } x^2 = 4ay$$

Points of intersection of these parabolas are obtained by solving these equations



$$y^2 = 4ax \rightarrow (1) \quad x^2 = 4ay \rightarrow (2)$$

From (2) $y = \frac{x^2}{4a}$

Using in (1) $\left(\frac{x^2}{4a}\right)^2 = 4ax$

$$\Rightarrow \frac{x^4}{16a^2} = 4ax \Rightarrow x^4 = 64a^3x$$

$$\Rightarrow x^4 - 64a^3x = 0 \Rightarrow x(x^3 - 64a^3) = 0$$

$$\Rightarrow x = 0, \quad x^3 = 64a^3 \text{ i.e. } x = 4a$$

when $x = 0$, $y = 0$ and when $x = 4a$, $y = 4a$

∴ Points of intersection are $O(0,0)$ and $A(4a,4a)$

From eq(1) $y = 2\sqrt{ax}$, and from eq(2) $y = \frac{x^2}{4a}$

Required area is given by

$$\begin{aligned}
 A &= \int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a} \right) dx \\
 &= 2\sqrt{a} \left(\frac{x^{3/2}}{3/2} \right)_0^{4a} - \frac{1}{4a} \left(\frac{x^3}{3} \right)_0^{4a} \\
 &= 4\sqrt{a} \frac{(4a)^{3/2}}{3} - \frac{1}{4a} \frac{(4a)^3}{3} \\
 &= \frac{4}{3} \sqrt{a} (4a) \sqrt{4a} - \frac{(4a)^2}{3} \\
 &= \frac{8}{3} \times 4a^2 - \frac{16a^2}{3} = \frac{32a^2 - 16a^2}{3} \\
 &= \frac{16}{3} a^2
 \end{aligned}$$

(4) Find the area bounded by the parabola

$y = 2x - x^2$ and the x-axis

$$y = 2x - x^2 \Rightarrow x^2 - 2x = -4$$

$$\text{i.e. } x^2 - 2x + 1 = 1 - 4$$

$$\text{i.e. } (x-1)^2 = -(4-1)$$

It is a parabola with vertex $(1,1)$ and axis $x-1=0$

Its intersection with x-axis is obtained by

Putting $y=0$ i.e. $x^2 - 2x = 0 \Rightarrow x(x-2) = 0$

i.e. $x=0$, $x=2$. The parabola meets the x-axis at $(0,0)$ and $(2,0)$

$$\text{Required area} = \int_0^2 y \, dx = \int_0^2 (2x - x^2) \, dx$$

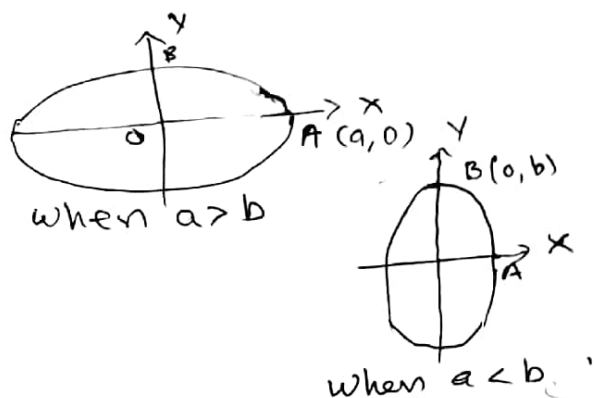
$$= \left(x^2 - \frac{x^3}{3} \right)_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3} \text{ sq. units.}$$

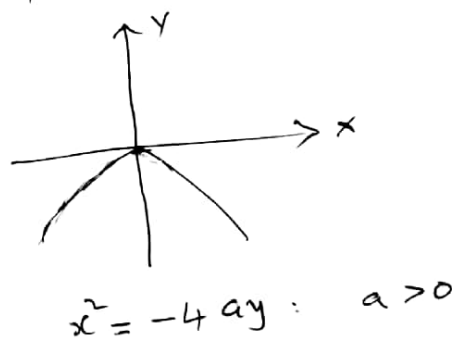
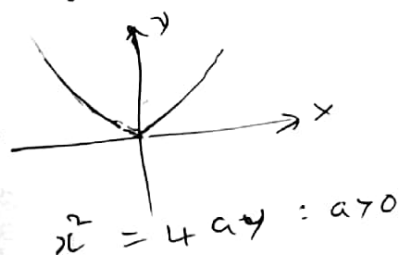
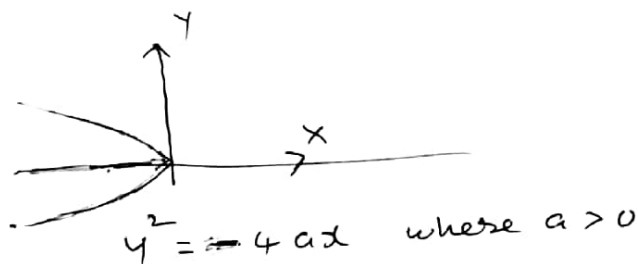
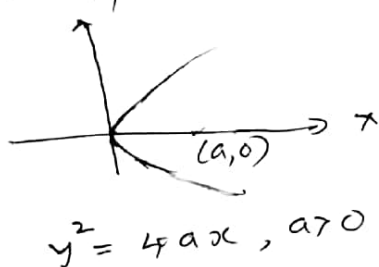


It is always necessary to trace the curve to determine the region whose area to be found. Some curves which occur in this module are given below

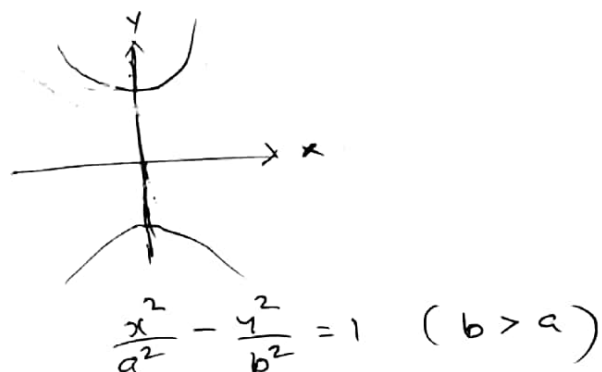
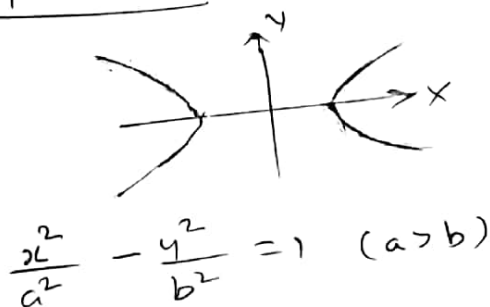
Ellipse : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



Parabolas:



Hyperbola



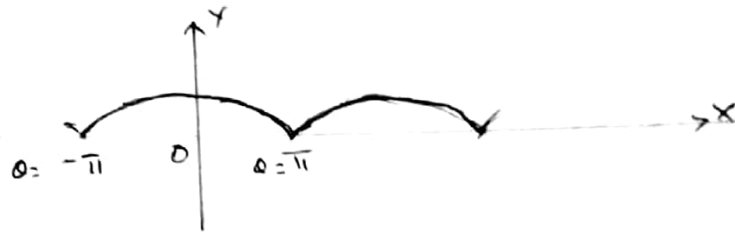
cycloid A cycloid is a curve traced by a point on a moving wheel.

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta)$$



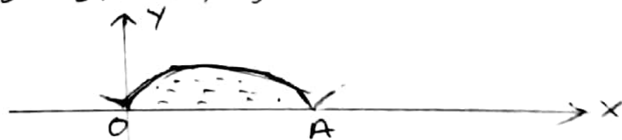
An arch of the cycloid is the portion between two consecutive contacts of the curve with its base, namely the x -axis. This curve consists of infinitely many arches. The curve meets the x -axis when $y=0$ i.e. when $a(1-\cos\theta)=0$
 $\cos\theta=1 \Rightarrow \theta=2n\pi, n=0,1,2,\dots$

$\theta=0$ and $\theta=2\pi$ are two consecutive points of intersection. i.e. one arch is between $\theta=0$ and $\theta=2\pi$



$x = a(\theta + \sin\theta), y = a(1 + \cos\theta)$ also represents a cycloid. One arch of this cycloid is between $\theta = -\pi$ and $\theta = \pi$.

- 5) Find the area enclosed by one arch of the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$



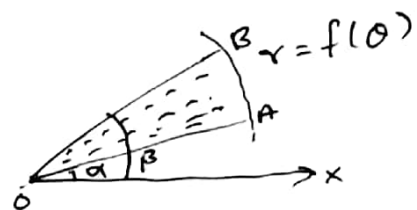
One arch of this curve is between $\theta=0$ and $\theta=2\pi$

$$\begin{aligned} \text{Required area} &= \int_{\theta=0}^{\theta=2\pi} y \, dx = \int_0^{2\pi} y \frac{dx}{d\theta} d\theta \\ &= \int_0^{2\pi} a(1-\cos\theta) \cdot a(1-\cos\theta) d\theta \\ &= a^2 \int_0^{2\pi} (1-\cos\theta)^2 d\theta \\ &= a^2 \int_0^{2\pi} (1 - 2\cos\theta + \cos^2\theta) d\theta \end{aligned}$$

$$\begin{aligned}
 &= a^2 \int_0^{2\pi} \left[1 - 2\cos\theta + \frac{1+\cos 2\theta}{2} \right] d\theta \\
 &= a^2 \left[\theta - 2\sin\theta + \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] \right]_0^{2\pi} \\
 &= a^2 \left[2\pi - 0 + \frac{2\pi}{2} \right] = a^2 (3\pi) \\
 &= 3\pi a^2
 \end{aligned}$$

- (6) Find the area of one arch of the cycloid
 $x = a(\theta + \sin\theta)$, $y = a(1 + \cos\theta)$

Area in polar co-ordinates



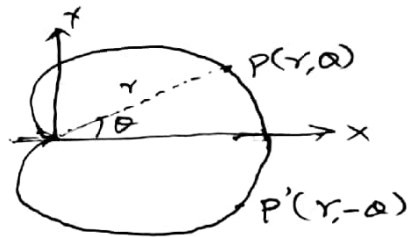
Area of the sector enclosed by the $r = f(\theta)$ and two radii vectors $\theta = \alpha$ and $\theta = \beta$ is given by

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

Example

- (1) Find the area enclosed by the cardioid
 $r = a(1 + \cos\theta)$

If $P(r, \theta)$ is a point on the given curve then $P'(r, -\theta)$ is also a point on it since $\cos(-\theta) = \cos\theta$.



$\therefore$ This curve is symmetric about the initial line. Therefore required area is twice the area above the initial line. For this part of the curve limits of θ are 0 and π .

$\therefore$ the required area = 2 $\times$ area of the cardioid above x-axis

$$\begin{aligned}
 &= 2 \int_0^{\pi} \frac{1}{2} r^2 d\theta \\
 &= a^2 \int_0^{\pi} (1 + \cos\theta)^2 d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= a^2 \int_0^{\pi} \left(2 \cos^2 \frac{\theta}{2}\right)^2 d\theta \\
 &= 4a^2 \int_0^{\pi} \cos^4 \frac{\theta}{2} d\theta \\
 &= 4a^2 \int_0^{\pi/2} \cos^4 \phi (2 d\phi) \\
 &= 8a^2 \int_0^{\pi/2} \cos^4 \phi d\phi \\
 &= 8a^2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\
 &= \frac{3\pi a^2}{2}
 \end{aligned}$$

$$\left[\begin{array}{l} \text{Put } \phi = \frac{\theta}{2} \\ d\phi = \frac{1}{2} d\theta \\ \text{when } \theta = 0, \phi = 0 \\ \text{when } \theta = \pi, \phi = \frac{\pi}{2} \end{array} \right]$$

Note $\int_0^{\pi/2} \sin^n \theta d\theta = \int_0^{\pi/2} \cos^n \theta d\theta = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{1}{2} \cdot \frac{\pi}{2}$
if n is even

$$\int_0^{\pi/2} \sin^n \theta d\theta = \int_0^{\pi/2} \cos^n \theta d\theta = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{2}{3} \times 1$$

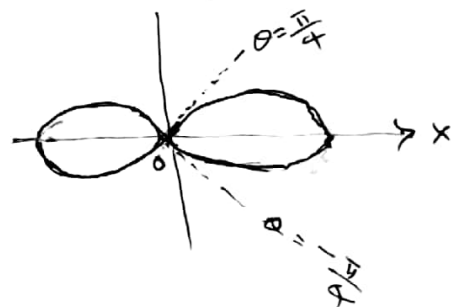
if n is odd

(2) Find the entire area of the Lemniscate of Bernoulli $r^2 = a^2 \cos 2\theta$

Since the given eq remains unchanged on changing θ to $-\theta$ or to $\pi - \theta$, the curve is symmetrical about both the axes.

when $r=0$, $\cos 2\theta = 0$ i.e. $2\theta = \pm \frac{\pi}{2}$
i.e. $\theta = \pm \frac{\pi}{4}$

The given curve consists of two loops, one of which lies between $\theta = -\frac{\pi}{4}$ and $\theta = \frac{\pi}{4}$



required area = 4 x area of one half loop above the initial line OX

$$\begin{aligned}
 \therefore A &= 4 \int_0^{\pi/4} \frac{1}{2} r^2 d\theta = 2a^2 \int_0^{\pi/4} \cos 2\theta d\theta = 2a^2 \left(\frac{\sin 2\theta}{2} \right)_0^{\pi/4} \\
 &= 2a^2 \times \frac{1}{2} (\sin \frac{\pi}{2} - \sin 0) = a^2 \times 1 = a^2
 \end{aligned}$$